

Inside STLC QA Skills Framework Roadmap

Core idea

The future QA career path is no longer:

Manual Tester → Automation Tester → Test Lead

It is becoming:

Foundation Tester → Automation Tester → Quality Engineer → AI-Enabled Quality Engineer → Quality Leader

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Level 1 — Foundation Tester

Goal

Build strong testing fundamentals before chasing tools.

Who this is for

Beginners, junior testers, career changers, manual testers, graduates, and people trying to get their first QA role.

Core skills

- Testing principles
- SDLC and STLC understanding
- Test case design
- Defect reporting
- Regression testing
- Exploratory testing basics
- Requirements analysis

- Communication with developers and stakeholders

Evidence to build

- Sample test cases
- Bug reports
- Test summary report
- Requirement review notes
- Exploratory testing charter

Level 2 — Automation Tester

Goal

Move from purely manual execution to technical testing support.

Who this is for

Manual testers wanting to progress, junior testers building technical confidence, or teams starting automation.

Core skills

- Test automation fundamentals
- Playwright, Cypress or Selenium basics
- API testing
- Postman
- Basic JavaScript or Python
- Version control basics
- CI/CD awareness
- Test data setup
- Automation maintenance

Evidence to build

- Automated smoke test suite
- API test collection
- GitHub repo

- Basic CI pipeline
- Simple automation framework
- Before/after automation improvement example

Level 3 — Quality Engineer

Goal

Move from testing tickets to improving quality across delivery.

Who this is for

Mid-level testers, automation testers, QA engineers, test leads, and teams moving towards quality engineering.

Core skills

- Shift-left testing
- Risk-based testing
- Test strategy
- Quality gates
- Agile delivery
- DevOps collaboration
- Root cause analysis
- Metrics and reporting
- Coaching developers on testability
- Improving release confidence

Evidence to build

- Risk-based test strategy
- Quality dashboard
- Release readiness checklist
- Defect trend analysis
- Root cause improvement plan
- Example of reducing regression effort

- Example of improving team quality process

Level 4 — AI-Enabled Quality Engineer

Goal

Use AI to accelerate testing, analysis, automation and quality decision-making.

Who this is for

Testers who want to future-proof their career, automation engineers, QA leads, and teams adopting AI.

Core skills

- AI prompting for QA
- AI-assisted test case generation
- AI-assisted requirements analysis
- AI-assisted automation support
- Log and defect analysis with AI
- Synthetic test data generation
- Test coverage analysis
- AI risk awareness
- LLM testing basics
- Human review and validation

Evidence to build

- AI-generated test scenarios improved by human review
- Prompt library for QA tasks
- AI-assisted requirements gap analysis
- AI-assisted bug investigation example
- Before/after productivity case study
- AI testing workflow diagram
- Ethical/risk review checklist

Level 5 — Quality Leader

Goal

Lead quality strategy, capability growth and transformation across teams.

Who this is for

QA Leads, Test Managers, Heads of QA, Engineering Managers, Delivery Leaders, CTOs, and senior testers moving into leadership.

Core skills

- Quality strategy
- Operating model design
- Team capability assessment
- Coaching and mentoring
- Tooling decisions
- AI adoption strategy
- Stakeholder influence
- Risk ownership
- Budget and ROI thinking
- Building quality culture

Evidence to build

- Team capability matrix
- Quality transformation roadmap
- Automation strategy
- AI adoption plan
- Training plan
- Quality maturity assessment
- Business case for quality investment
- Executive-level reporting

The 12-Month Roadmap

Months 1–3: Build the Foundation

Focus on:

- Testing fundamentals
- Test design
- Defect reporting
- Requirements analysis
- Exploratory testing
- Communication

Outcome:

You can test properly before relying on tools.

Months 4–6: Add Technical Capability

Focus on:

- API testing
- Postman
- Basic automation
- JavaScript or Python basics
- Git basics
- CI/CD awareness

Outcome:

You can support modern engineering teams, not just execute manual tests.

Months 7–9: Become AI-Enabled

Focus on:

- AI prompting for QA
- AI-assisted test design
- AI-assisted requirements analysis

- AI-assisted automation support
- AI-assisted defect analysis
- Building a personal QA prompt library

Outcome:

You can use AI to speed up repetitive work while improving your thinking time.

Months 10–12: Move Towards Quality Engineering

Focus on:

- Risk-based testing
- Test strategy
- Quality metrics
- Root cause analysis
- Release confidence
- Stakeholder communication
- Business impact

Outcome:

You stop being seen as someone who only tests software and start being seen as someone who improves quality.

All these courses can be found on Inside STLC Academy

